Practice workbook · Emerging tech: quantum, DNA computing, AI and nanotechnology

Definitions, characteristics and applications of quantum computing, DNA computing, artificial intelligence and nanotechnology, including discussing AI's impact.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-17

1. Question 2 · part d · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part d only.

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2 Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

(2)

1 .....

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2 .....

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(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

(1)

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

(2)

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1. Question 2 · part d · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part d only.

(c) The CPU performs the fetch-decode-execute cycle.

(i) Give **three** factors that affect the performance of the CPU.

(3)

1 .....

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2 .....

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3 .....

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(ii) Describe how the address bus is used during the fetch-decode-execute cycle.

(2)

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(d) Classical computers store data using bits that exist in one of two binary states, 1 or 0.

Quantum computers store data in a superposition of states.

Give **two other** characteristics of quantum computers.

(2)

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(Total for Question 2 = 12 marks)

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1. Question 2 · part d · 2 marks · Working and answer

Working and answer

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2. Question 6 · part c · 6 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

6 A train company uses ticket vending machines at each station.

(a) The machines use embedded systems.

(i) Explain **one** benefit of using an embedded system in these machines.

(2)

(ii) Customers use a touch screen to select their destination. They can pay by cash or bank card. Their tickets and a receipt are printed.

The touch screen is controlled by an embedded system.

Give **two** other hardware components in the ticket machine that are controlled by embedded systems.

(2)

1 .....

2 .....

(iii) The ticket machine uses data encryption when a customer pays using a bank card.

State why data encryption is used in this case.

(1)

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2. Question 6 · part c · 6 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

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(b) Compare **four** features of high-level and low-level programming languages.

(4)

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2. Question 6 · part c · 6 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

(c) Artificial intelligence (AI), in many forms, has an increasing impact on our lives.

Discuss this statement considering characteristics, uses and ethical issues.

(6)

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(Total for Question 6 = 15 marks)

2. Question 6 · part c · 6 marks · Working and answer

Working and answer

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3. Question 6 · part a.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a.i only.

6 (a) Two emerging technologies are DNA computing and quantum computing.

(i) Explain **one** difference between DNA computing and traditional computing.

(2)

(ii) Give a definition of the term **superposition** in quantum computing.

(1)

(b) Ethically, everyone should be able to access the benefits of technology.

Give **two** reasons why this may not be possible.

(2)

1

2

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3. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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4. Question 6 · part a.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a.ii only.

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(2)

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(2)

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4. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 5 · part a.i · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a.i only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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5. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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6. Question 4 · part d.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d.i only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

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6. Question 4 · part d.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d.i only.

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(d) Marie is a researcher in nanotechnology.

(i) State what is meant by the term **nanotechnology**.

(1)

(ii) Give **one** application of nanotechnology.

(1)

(e) Marie has a mobile phone.

The address of the device is:

2024:0eb8:3c4d:0015:0000:0000:1a2f:1a2b

Describe the relationship between the structure of this address and the total number of bits required to store it.

(3)

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6. Question 4 · part d.i · 1 mark · Working and answer

Working and answer

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7. Question 4 · part d.ii · 1 mark

2024 JUN · Paper 01

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7. Question 4 · part d.ii · 1 mark

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Further practice · Attempt part d.ii only.

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7. Question 4 · part d.ii · 1 mark · Working and answer

Working and answer

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